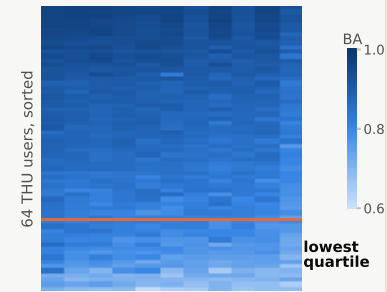
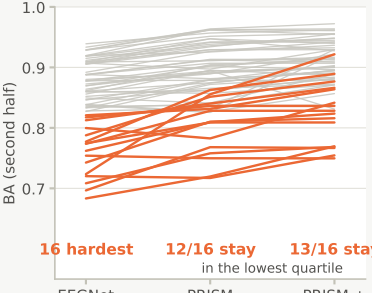
Calibration-free RSVP-EEG decoding: difficulty is bound to the user, not the decoder

1 The same users are hardest for every decoder



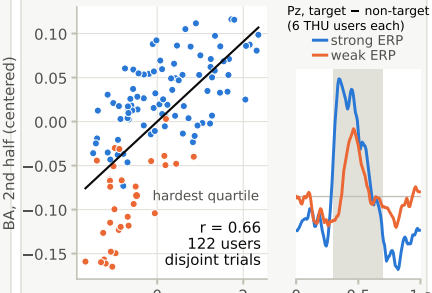
Participants explain 77–94% of BA variance, decoders only 2–11% (3 datasets)

2 Better decoders and calibration do not rescue them



Source alignment: no consistent gain.
Own-data calibration: only +0.016 BA

3 Weak ERP strength explains why



→ Screen new users with ~40 target trials (2–7 min)

11 decoders · 3 source-alignment objectives · 3 public RSVP datasets · 122 participants · no data from the held-out user